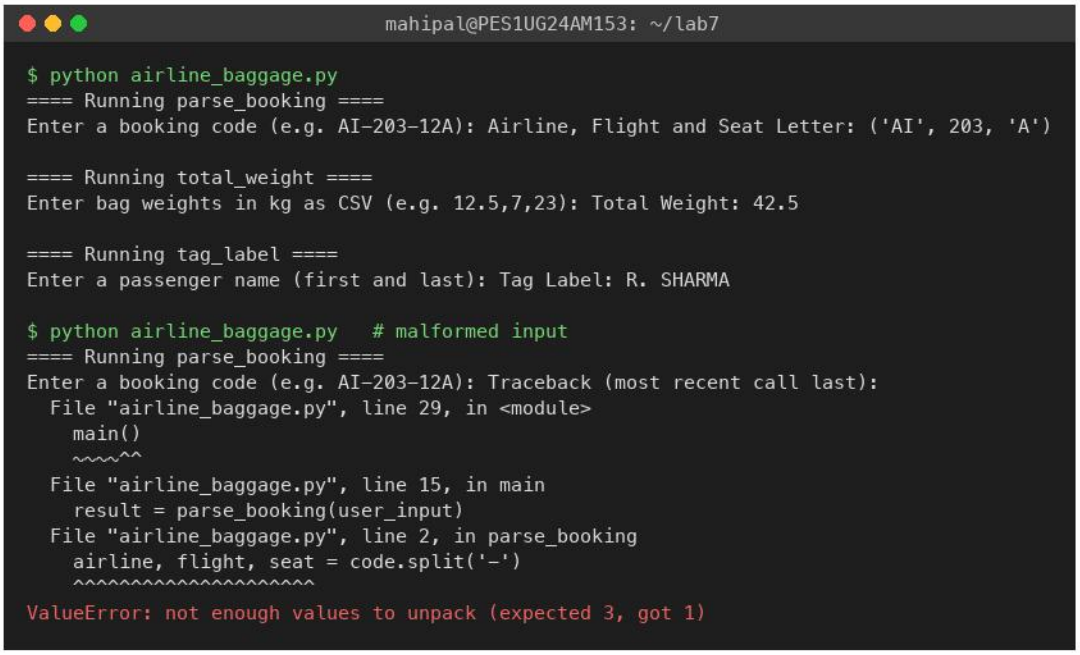
Lab 7: Fuzz Testing with Hypothesis

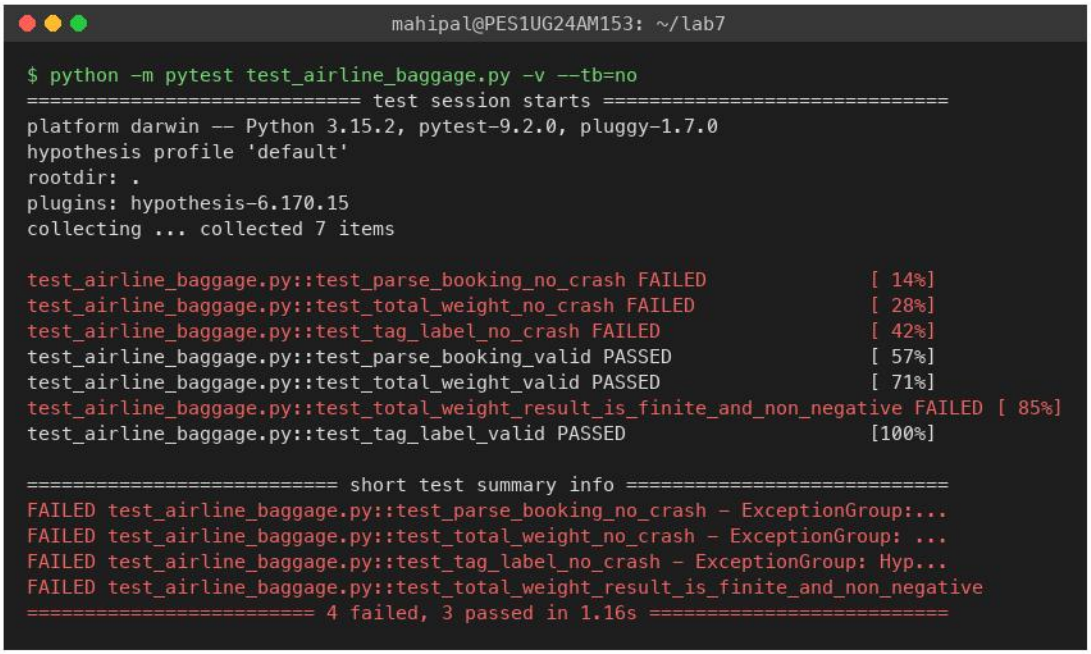
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Repo: SETAPESU26/L6_L7_28_airline_baggage (lab7/airline_baggage.py)

Step 2: Running the buggy airline_baggage.py (valid input works, malformed input crashes)



Step 4: Hypothesis fuzz tests failing on the buggy code



Step 4: Minimal failing inputs found by Hypothesis

```
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$ python -m pytest test_airline_baggage.py   # falsifying examples found by Hypothesis
ValueError: not enough values to unpack (expected 3, got 1)
AssertionError: parse_booking crashed with: ValueError('not enough values to unpack (expected 3, got 1)')
Failing test case: test_parse_booking_no_crash(
s='',
AttributeError: 'NoneType' object has no attribute 'split'
AssertionError: parse_booking crashed with: AttributeError("'NoneType' object has no attribute 'split'")
Failing test case: test_parse_booking_no_crash(
s=None,
ValueError: could not convert string to float: ''
AssertionError: total_weight crashed with: ValueError("could not convert string to float: ''")
Failing test case: test_total_weight_no_crash(
s='',
AttributeError: 'NoneType' object has no attribute 'split'
AssertionError: total_weight crashed with: AttributeError("'NoneType' object has no attribute 'split'")
Failing test case: test_total_weight_no_crash(
s=None,
IndexError: list index out of range
AssertionError: tag_label crashed with: IndexError('list index out of range')
Failing test case: test_tag_label_no_crash(
s='',
AttributeError: 'NoneType' object has no attribute 'split'
AssertionError: tag_label crashed with: AttributeError("'NoneType' object has no attribute 'split'")
Failing test case: test_tag_label_no_crash(
s=None,
ValueError: could not convert string to float: ''
Failing test case: test_total_weight_result_is_finite_and_non_negative(
s='',
AssertionError: assert (inf is None or (False))
Failing test case: test_total_weight_result_is_finite_and_non_negative(
s=inf,
```

Step 5: Running the fixed airline_baggage.py (bad input handled gracefully)

```
mahipal@PES1UG24AM153: ~/lab7

$ python airline_baggage.py
==== Running parse_booking ====
Enter a booking code (e.g. AI-203-12A): Airline, Flight and Seat Letter: ('AI', 203, 'A')

==== Running total_weight ====
Enter bag weights in kg as CSV (e.g. 12.5,7,23): Total Weight: 42.5

==== Running tag_label ====
Enter a passenger name (first and last): Tag Label: R. SHARMA

$ python airline_baggage.py   # malformed input
==== Running parse_booking ====
Enter a booking code (e.g. AI-203-12A): Invalid booking code. Expected format like AI-203-12A.

==== Running total_weight ====
Enter bag weights in kg as CSV (e.g. 12.5,7,23): Invalid weights. Enter non-negative numbers separated by commas.

==== Running tag_label ====
Enter a passenger name (first and last): Invalid name. Enter a first and last name.
```

Step 6: All fuzz tests passing on the fixed code

